

Sl. No	Reference Designator	Component Description	Value/Rating	Package Type	Brand/Manufacturer	Vendor/Source	Part Number	Total Qty	Rate(Rs)
1	D1, D2	0805 Surface Mount LED Red	30 mA max	0805, SMD	Generic	ROBU	nil	2	2.88
2	SW1, SW2	Tactile Switch SPST-NO Top Actuated Surface Mount	50 mA, 12V	4.5x4.5 mm SMD	C&K	DigiKey	PTS647SN38SMTR2 LFS	2	57.34
3	U1	ARM Cortex-M4 32-bit Microcontroller	84MHz 512 KB Flash 96 KB SRAM, 64pins	LQFP-64	STMicroelectronics	Evelta	STM32F401RET6	1	342.2
4	J1	Micro USB Connector B Female	3.3V	5 Pin SMD, mid mount 4 mounting THT	Molex	Sunrom Electronics	47642-0001	1	27.6
5	J2	Female Straight Header Strip	1×5, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7846	1	5.29
6	J3, J4	Female Straight Header Strip	1×4, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7845	2	4.72
7	J5	Female Straight Header Strip	1×6, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7847	1	5.75
8	J7	Female Straight Header Strip	1×10, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7850	1	7.59
9	U2	Fixed 3.3 V Low-Dropout Linear Voltage Regulator (LDO)	3.3V, 800mA	SOT-223-4	STMicroelectronics	ROBU	LD1117S33CTR	1	29
10	F1	PPTC Resettable Fuse, 1206	24V max, 500mA	1206 SMD	WEITE	Sharvi Electronics	WT1206-050	1	12.35
11	Y1	8MHz Surface Mount Crystal	8MHz, 20pF	HC-49S-SMD	YXC Crystal Oscillators	ROBU	X49SM8MSD2SC	1	13
12	R1, R2, R5	10K Ohm 1/8 Watt Resistor	10K Ohm ±1% tolerance	0805, SMD	YAGEO	Sharvi Electronics	RC0805FR-0710KL	3	2.46
13	R3	330 Ohm 1/8 Watt Resistor	330 Ohms Tolerance: ±1%	0805, SMD	YAGEO	Sharvi Electronics	RC0805FR-07330RL	1	0.86
14	R4	1K Ohm 1/8 Watt Resistor	1K Ohm Tolerance: ±1%	0805, SMD	Royal Ohm	Sharvi Electronics	0805S8F1001T5E	1	0.82
15	R6, R7	4.7K Ohm, 1/8 Watt Resistor	4.7K Ohm, ±1% tolerance	0805, SMD	Watts	Sharvi Electronics	CR-05FTP-4K7	2	1.64
16	C1, C2	10 µF Multilayer Ceramic Capacitor (MLCC)	10 µF, 16 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R8BB106	2	5.44
17	C5, C6, C7, C8, C9, C10, C14	100nF Multilayer Ceramic Capacitor (MLCC)	100nF, 50 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R9BB104	7	13.79
18	C11, C12	4.7 µF Multilayer Ceramic Capacitor (MLCC)	4.7 µF, 25 V, ±10%	0805, SMD	Walsin	Sharvi Electronics	0805X475K250CT	2	4.24
19	C13	1µF Multilayer Ceramic Capacitor (MLCC)	1 µF, 16 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R7BB105	1	2.15
20	C3, C4	22 pF Ceramic Capacitor, C0G/NP0	22 pF, 100 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805CRNP00BN221	2	4.5